

Machine-Failure Model Comparison

Table 1 – 5-fold CV on 80 % Training Data

ML Trained Model	Its Best Set of Parameter Values	MCC on 5-fold CV (80 % train)
MLP	{'clflearning_rate': 'constant', 'clfhidden_layer_sizes': (100, 50), 'clfalpha': 0.001, 'clfactivation': 'relu'}	0.8232
SVM	{'clfC': 100, 'clfkernel': 'rbf', 'clf__gamma': 'auto'}	0.8196
...	...	...

Table 2 – Performance on 20 % Hold-out Test Set

ML Trained Model	Its Best Set of Parameter Values	MCC on 20 % Test set
MLP	{'clflearning_rate': 'constant', 'clfhidden_layer_sizes': (100, 50), 'clfalpha': 0.001, 'clfactivation': 'relu'}	0.8239
SVM	{'clfC': 100, 'clfkernel': 'rbf', 'clf__gamma': 'auto'}	0.7945
...	...	...

Conclusion – Selected Model

The **MLP** achieved the highest MCC on both cross-validation and the held-out test set, so we choose it as the production model.